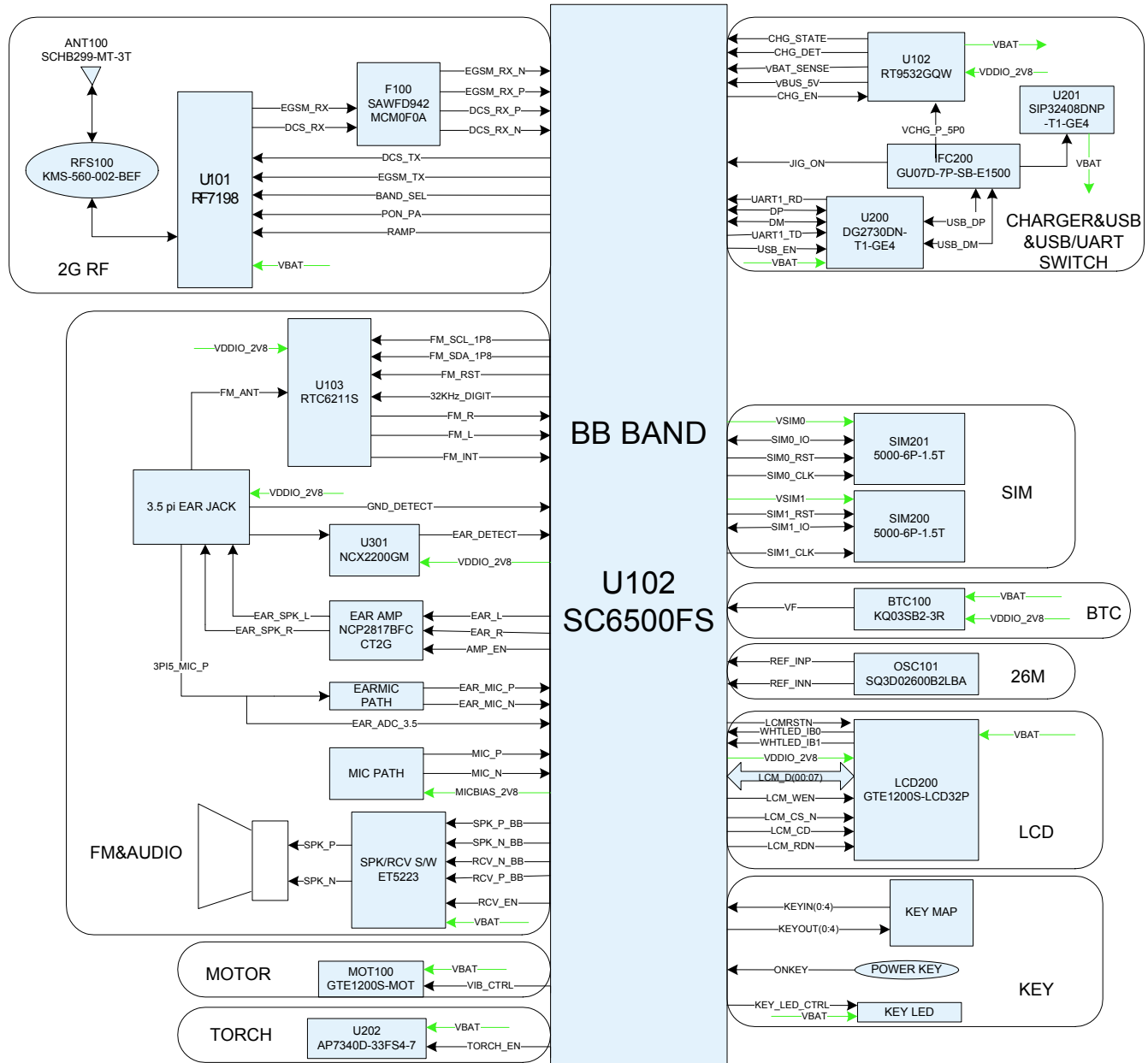


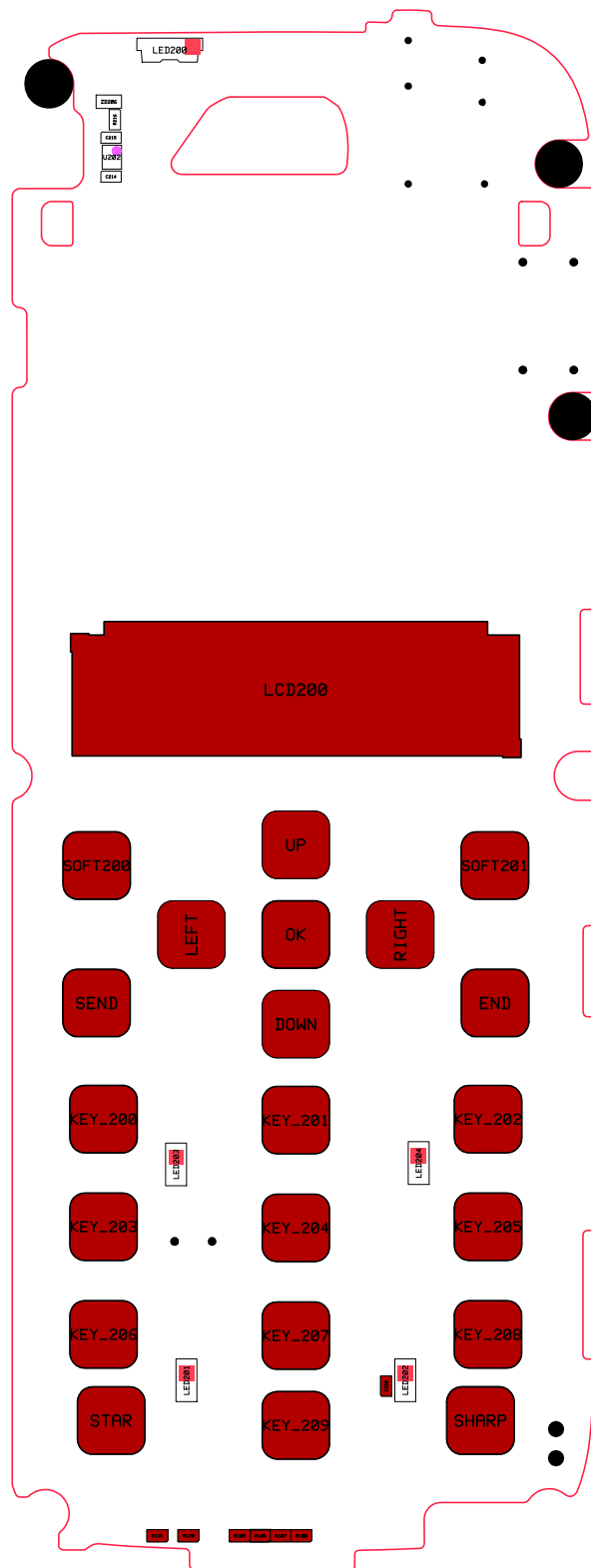
8. Level 3 Repair

8-1. Block Diagram

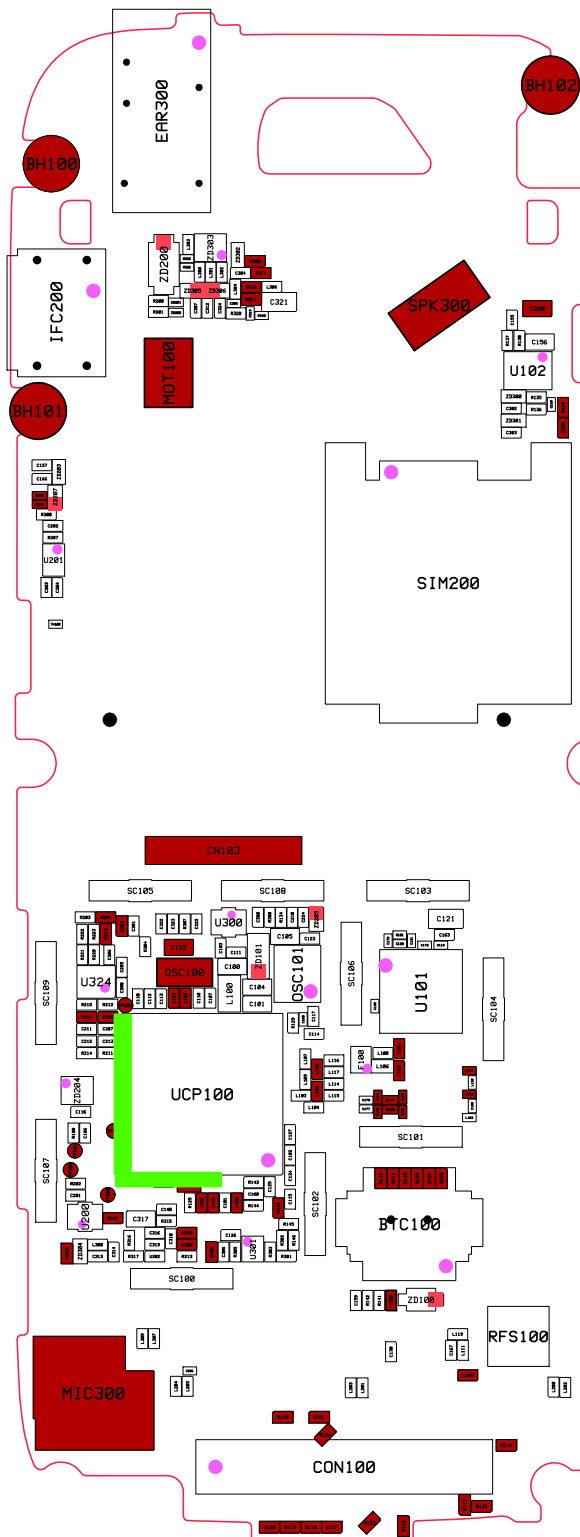


8-2. PCB Diagrams

8-2-1. Top



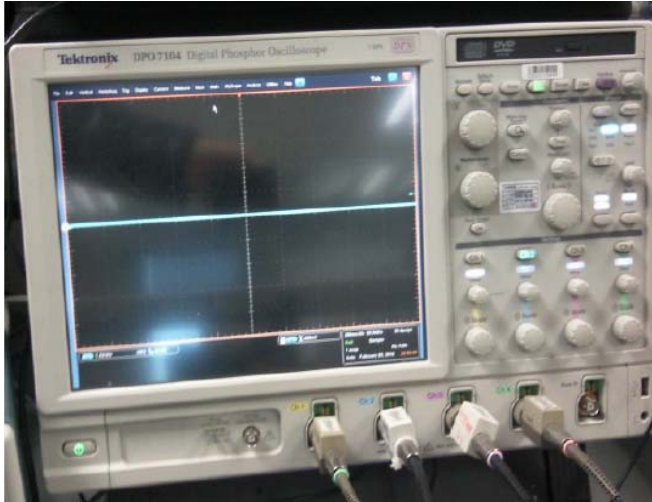
8-2-1. Bot



8-3. Flow Chart of Troubleshooting

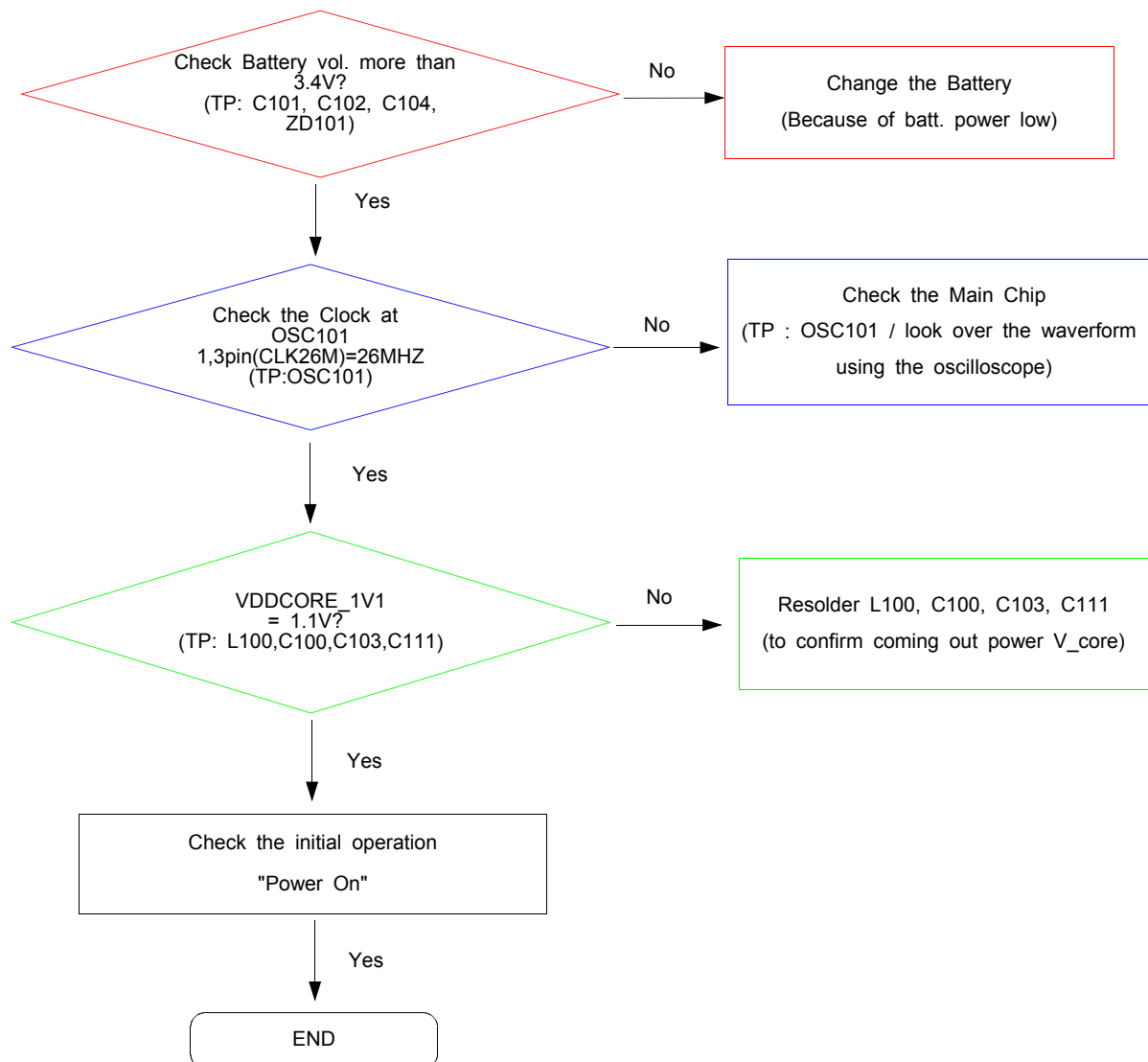
※ presetting methods for checking TP

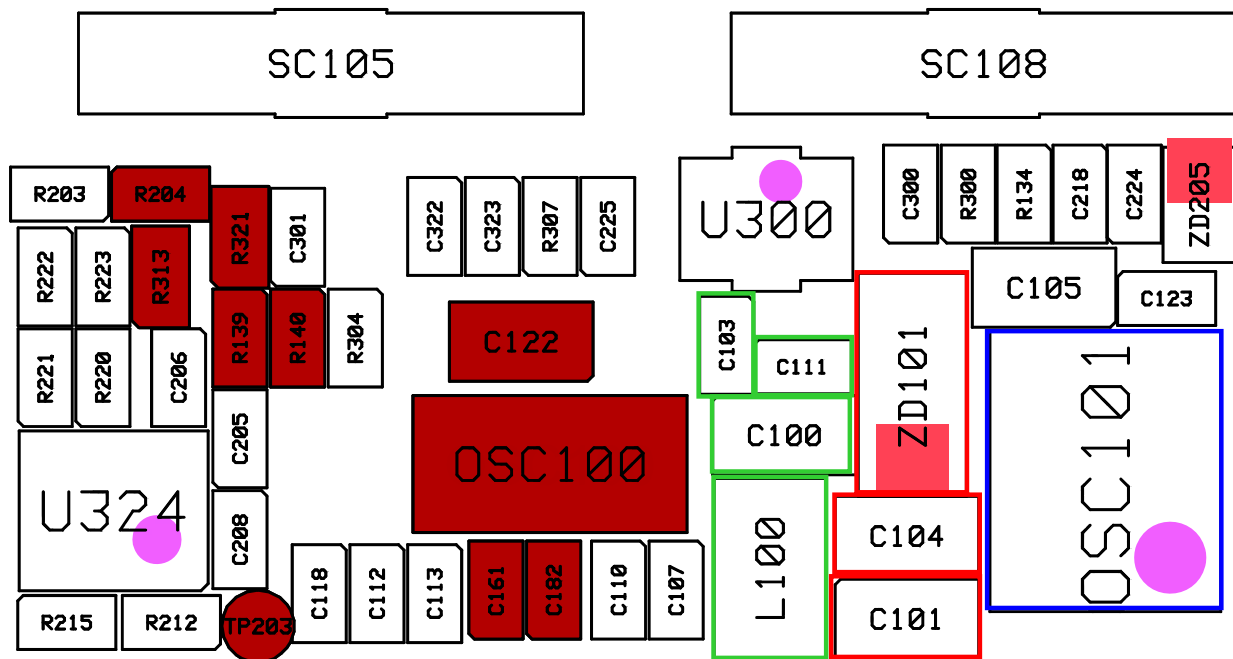
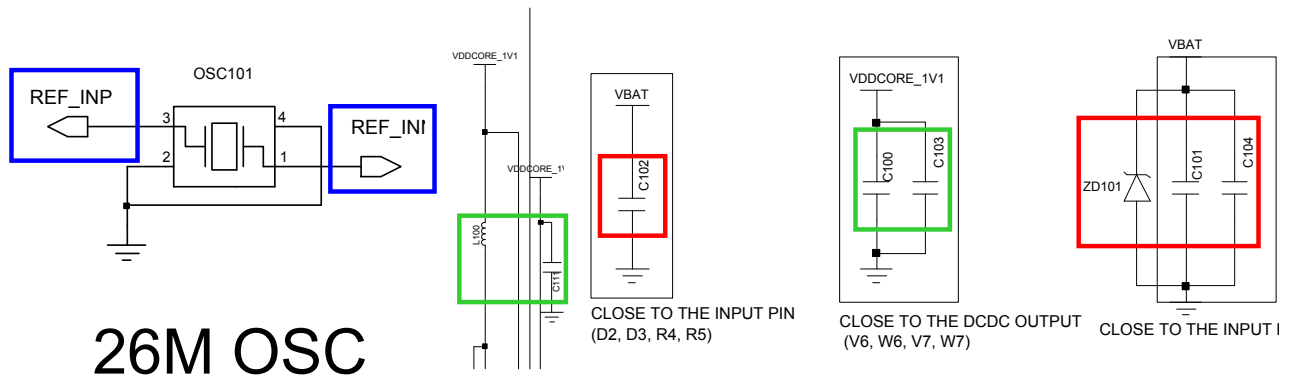
- GND & TP(exp. VBUS_5.0V= C155) using Oscilloscope
- look over the coming out signal.



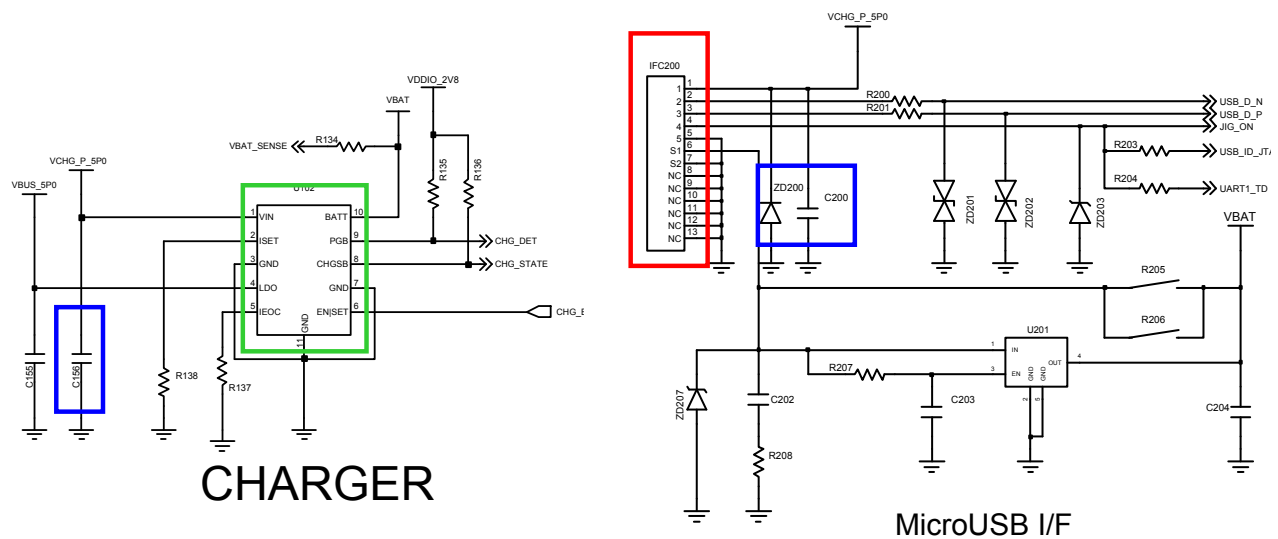
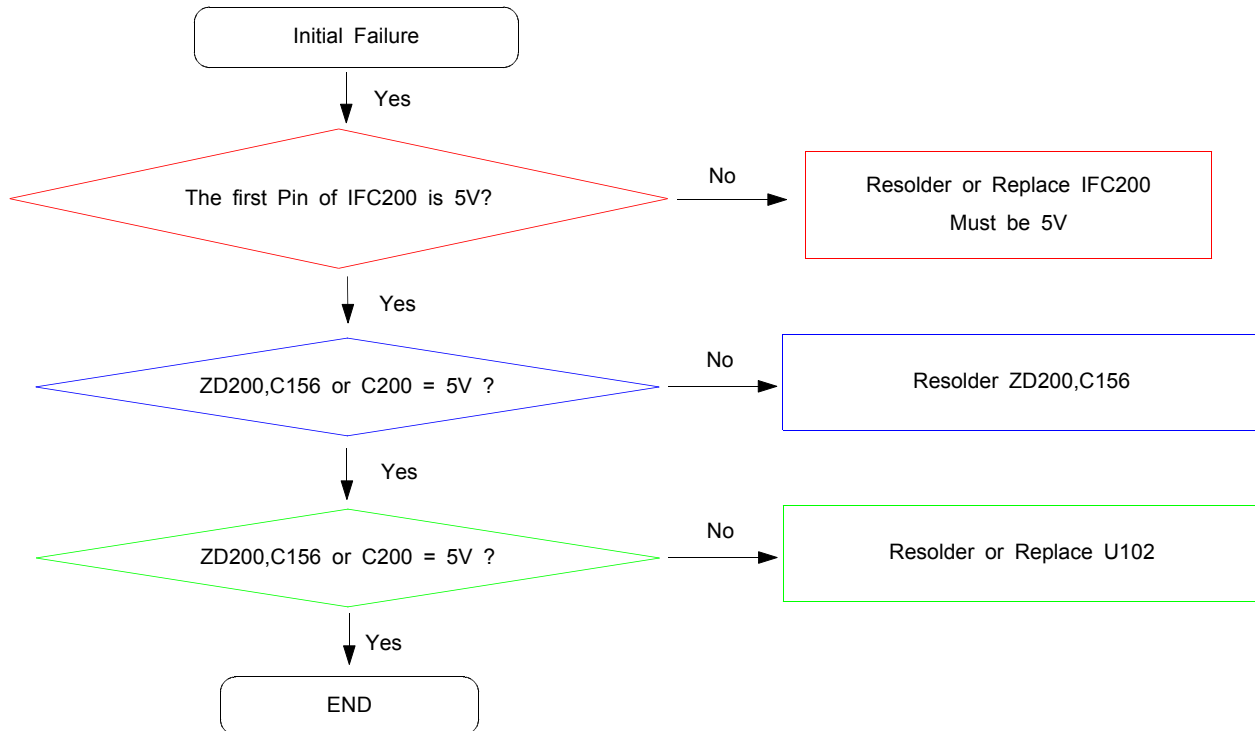
← Oscilloscope

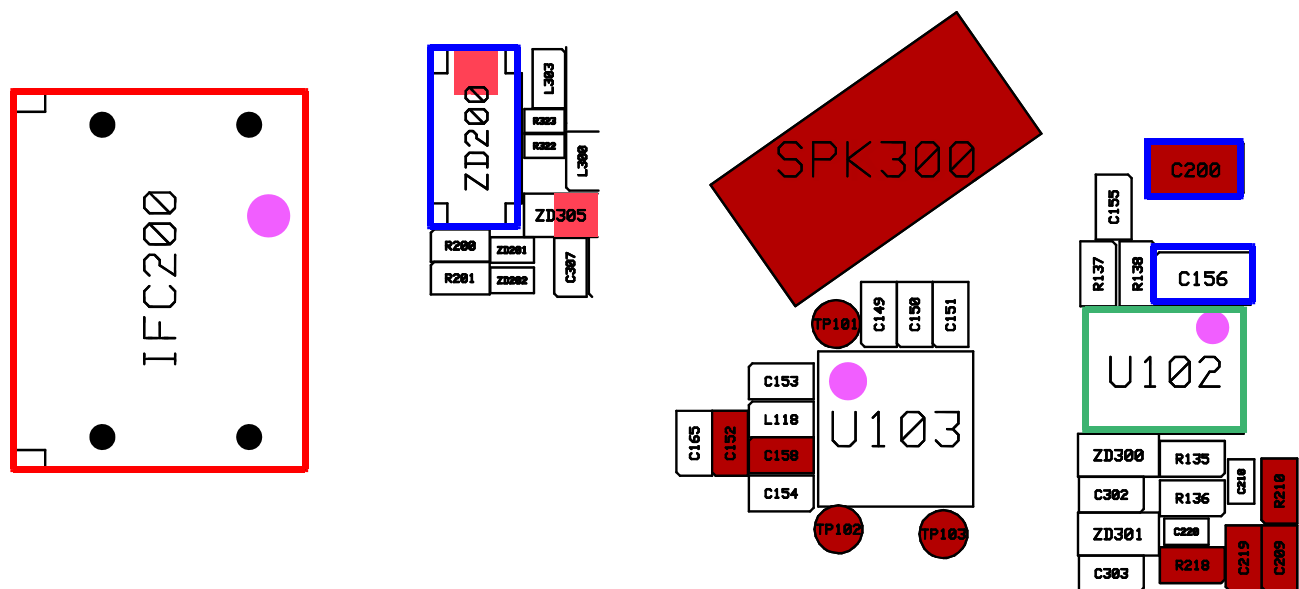
8-3-1. Power On



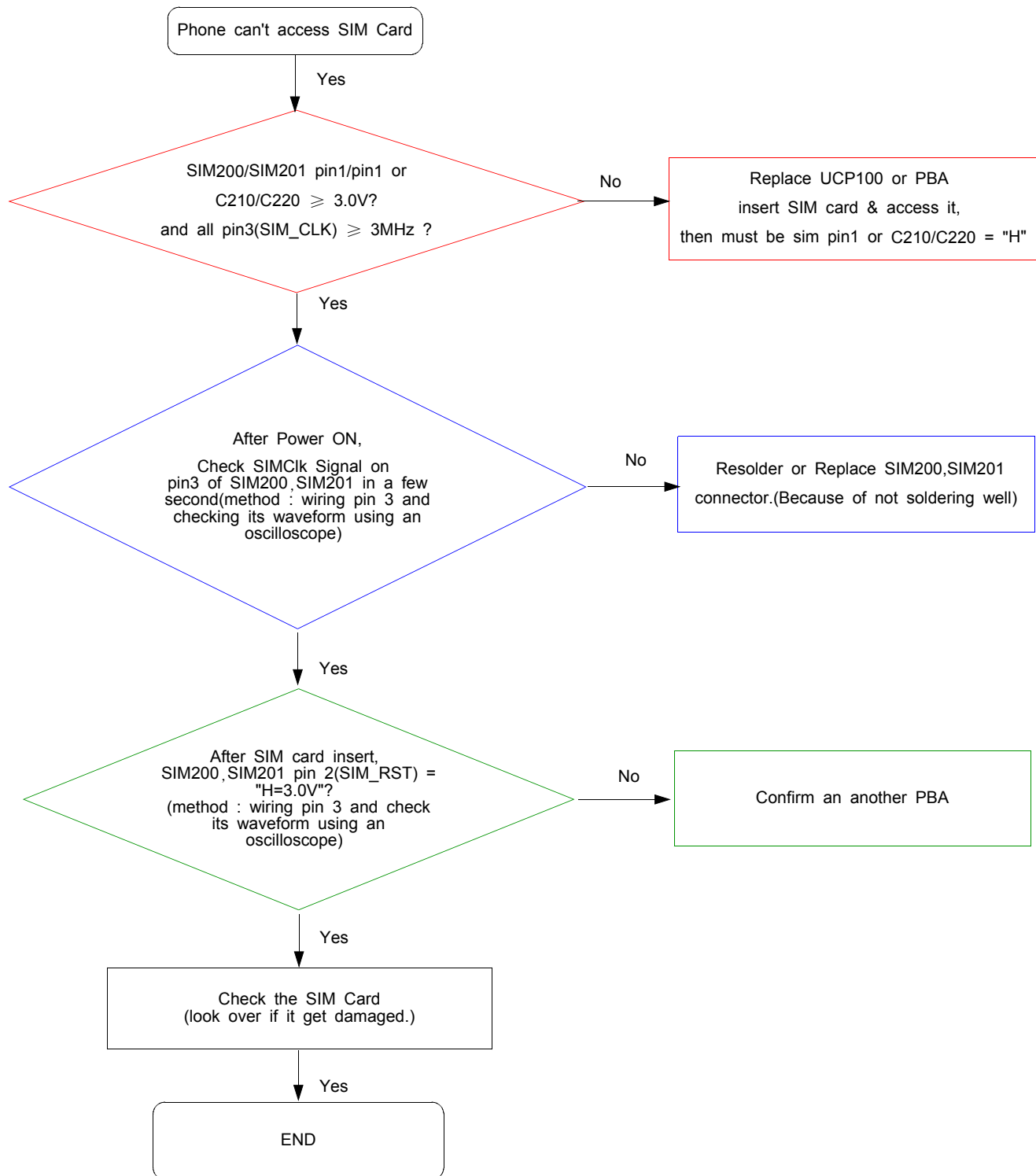


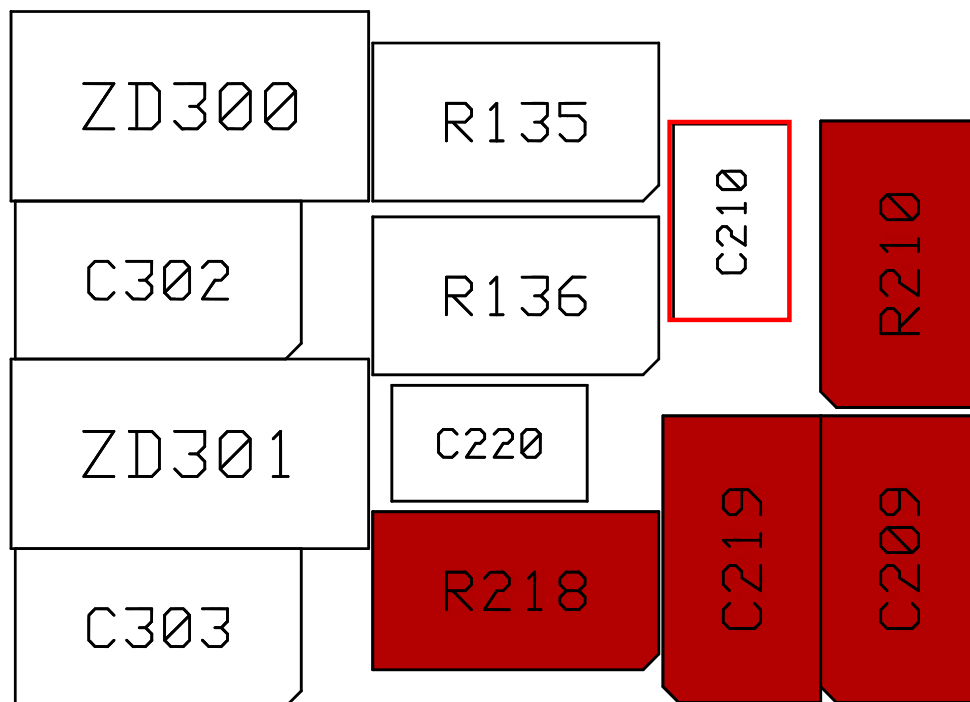
8-3-2. Charging Part



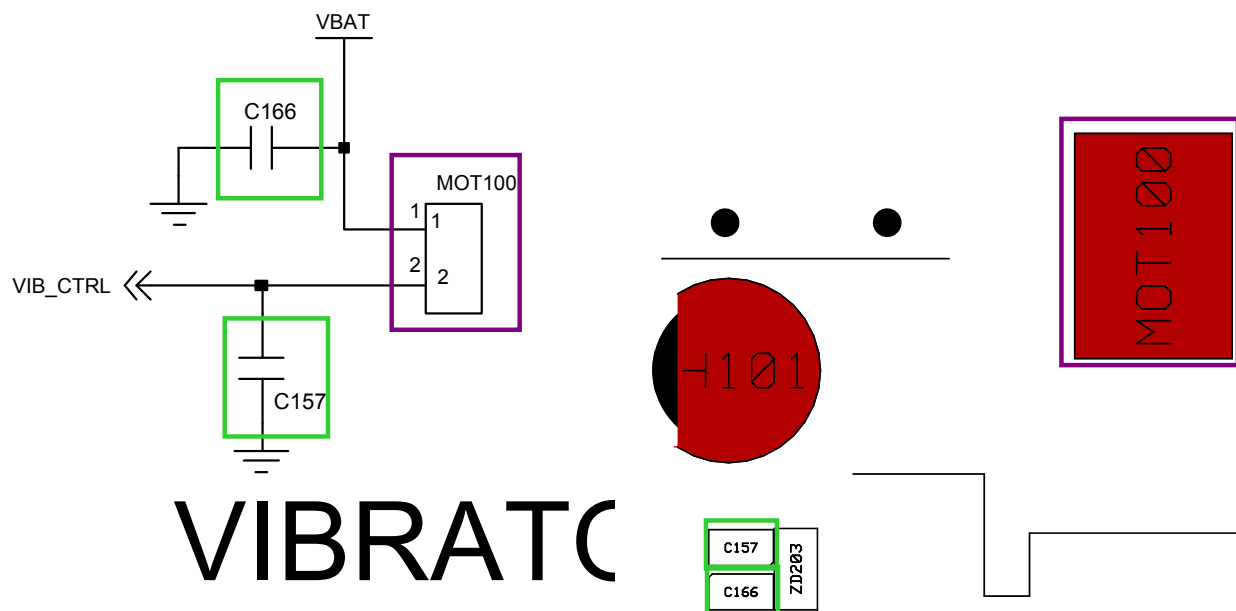
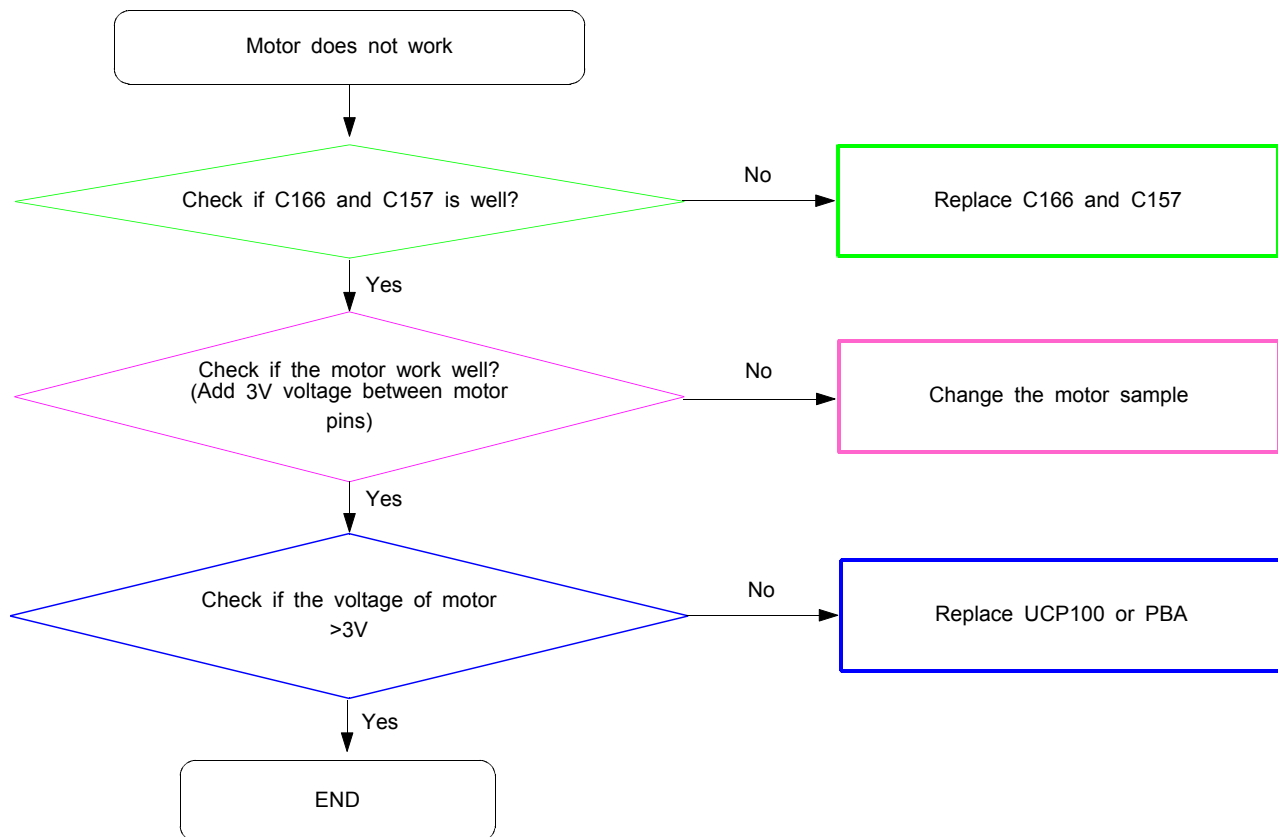


8-3-3. Sim Part

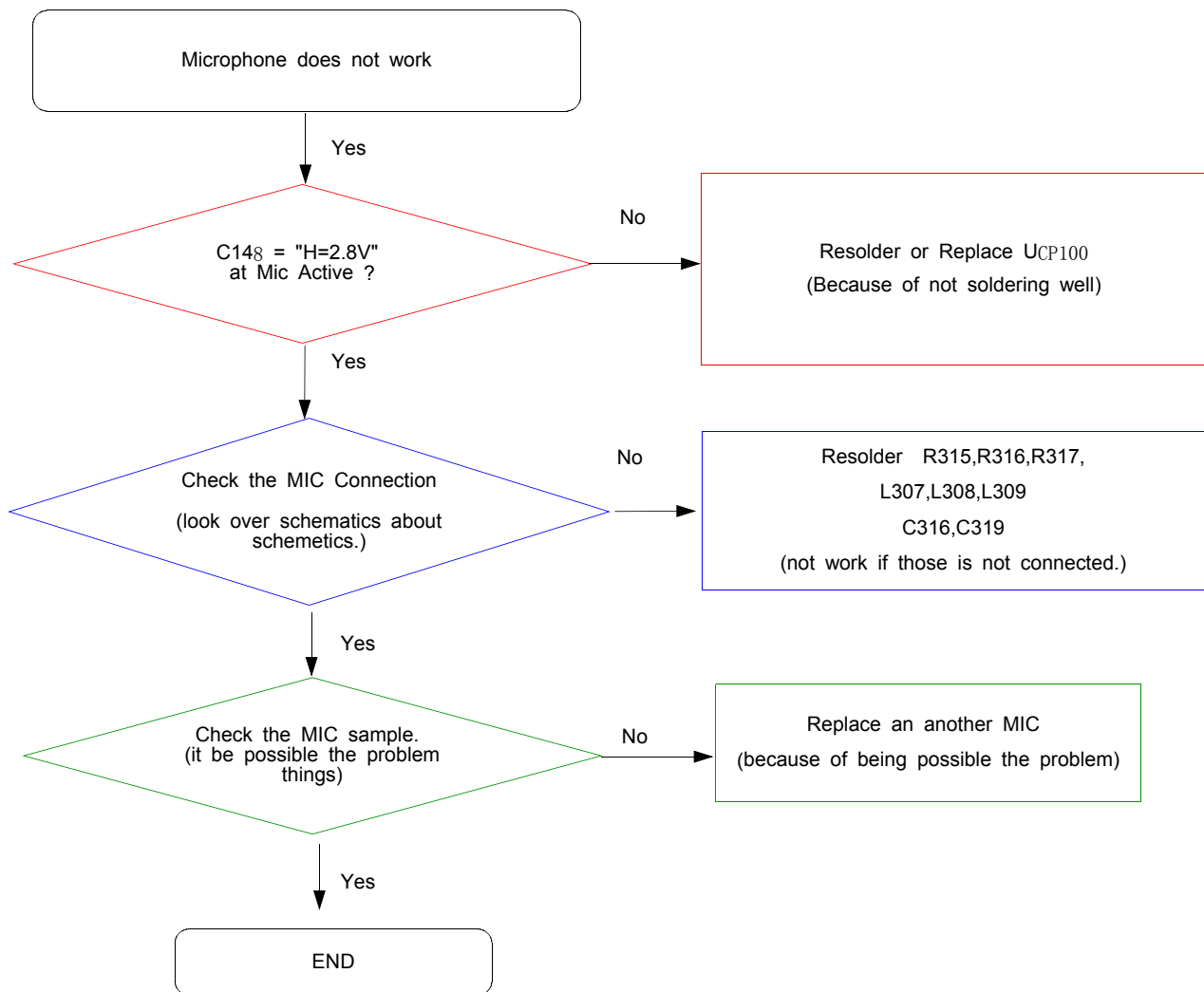


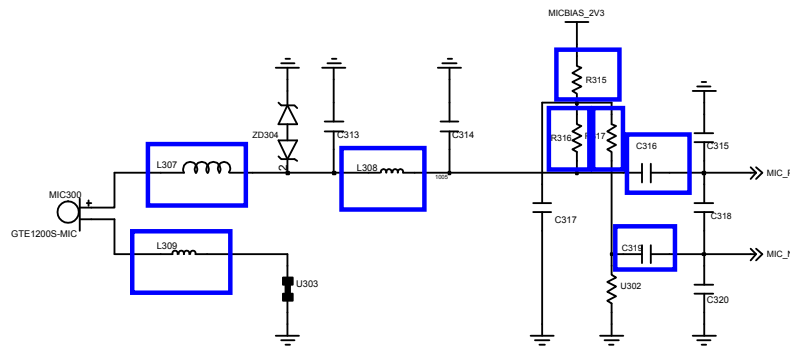
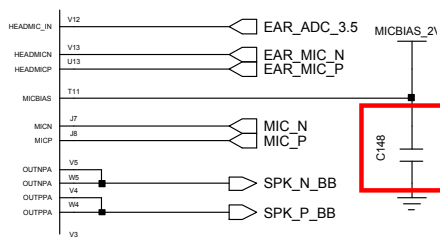


8-3-4. Motor Part

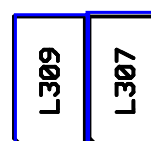
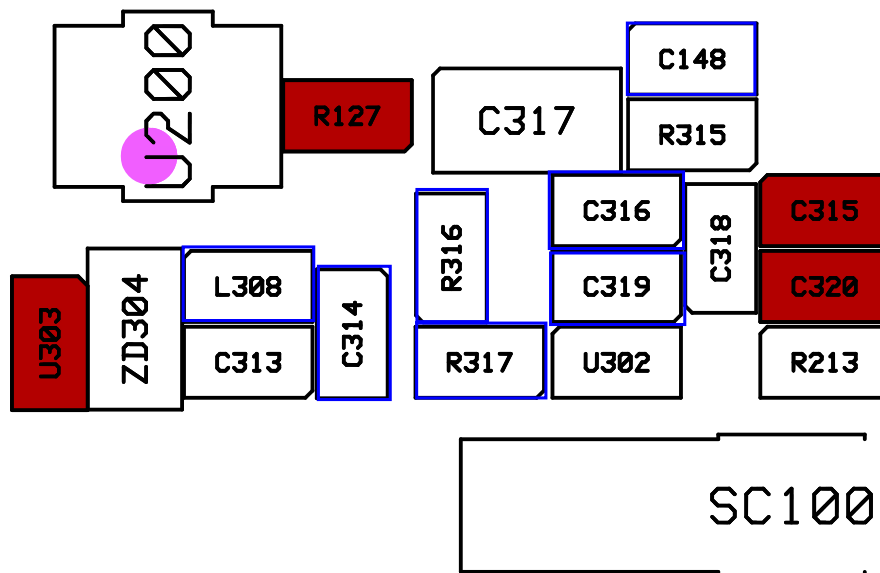


8-3-5. Microphone Part

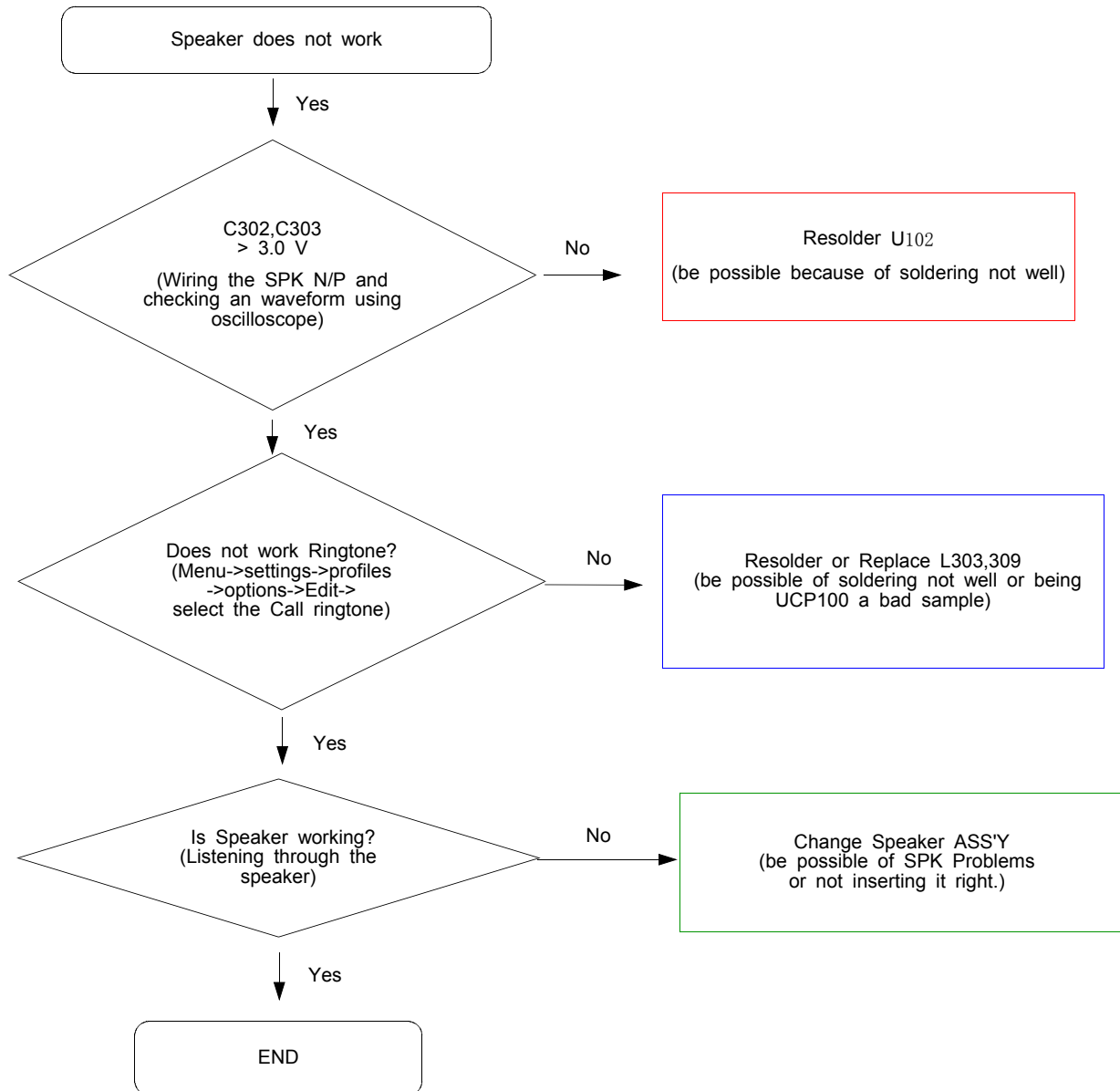


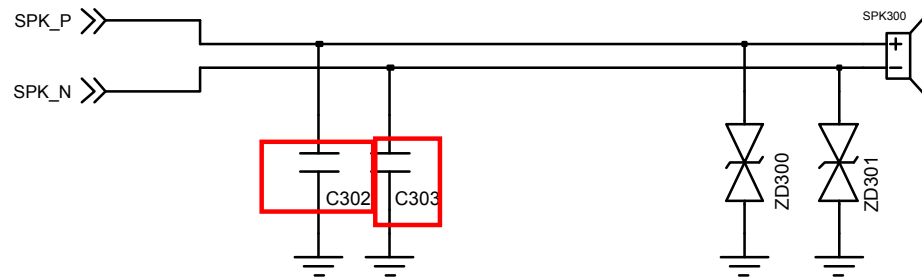


MAIN_MIC

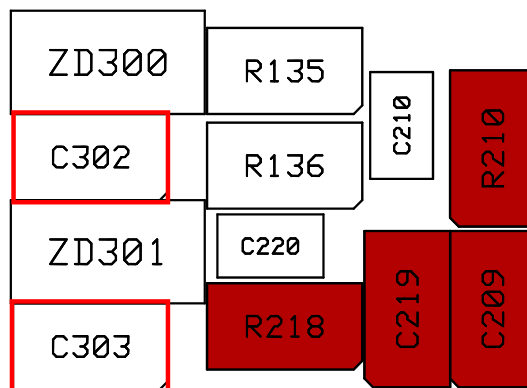


8-3-6. Speaker Part

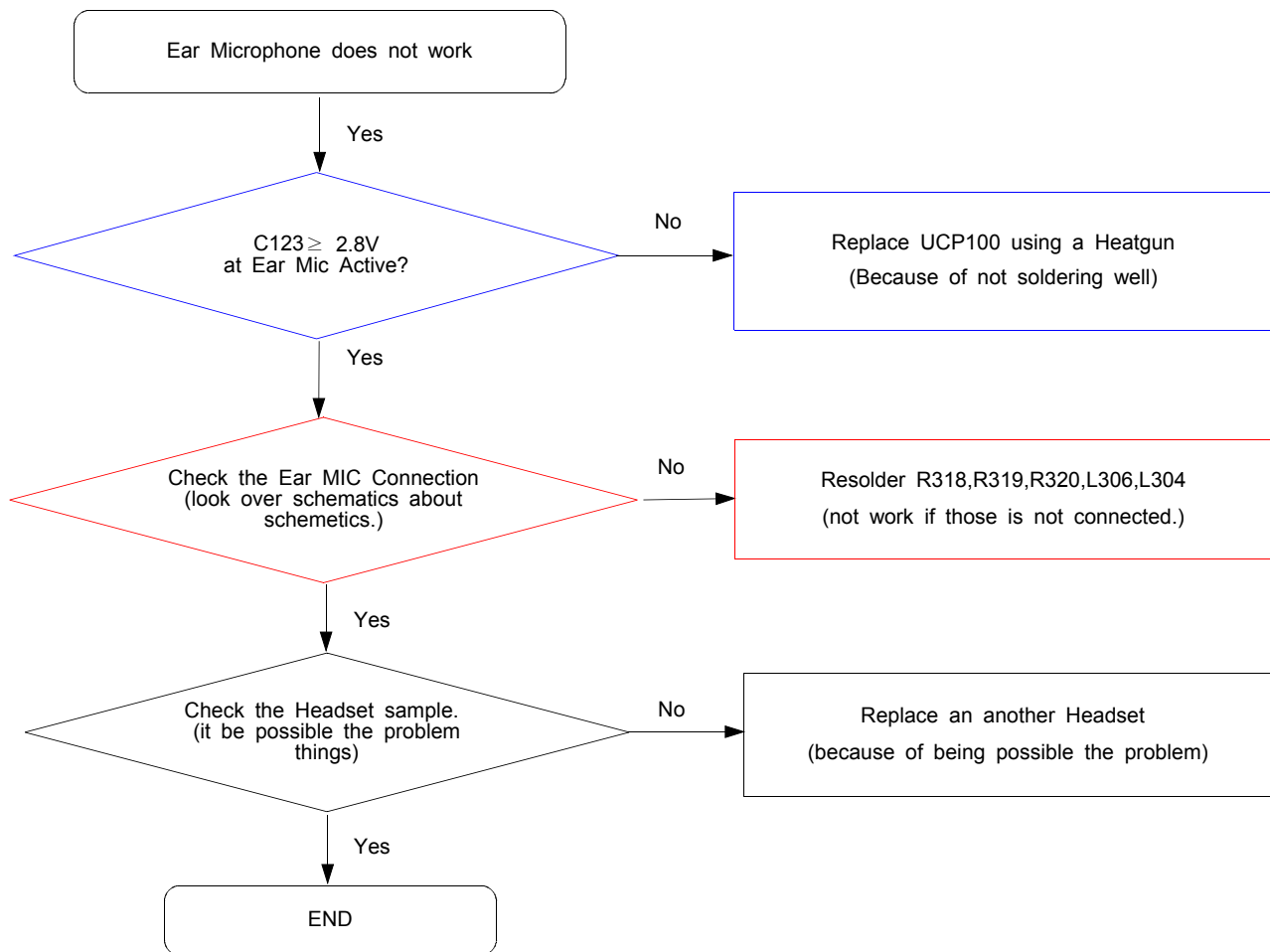


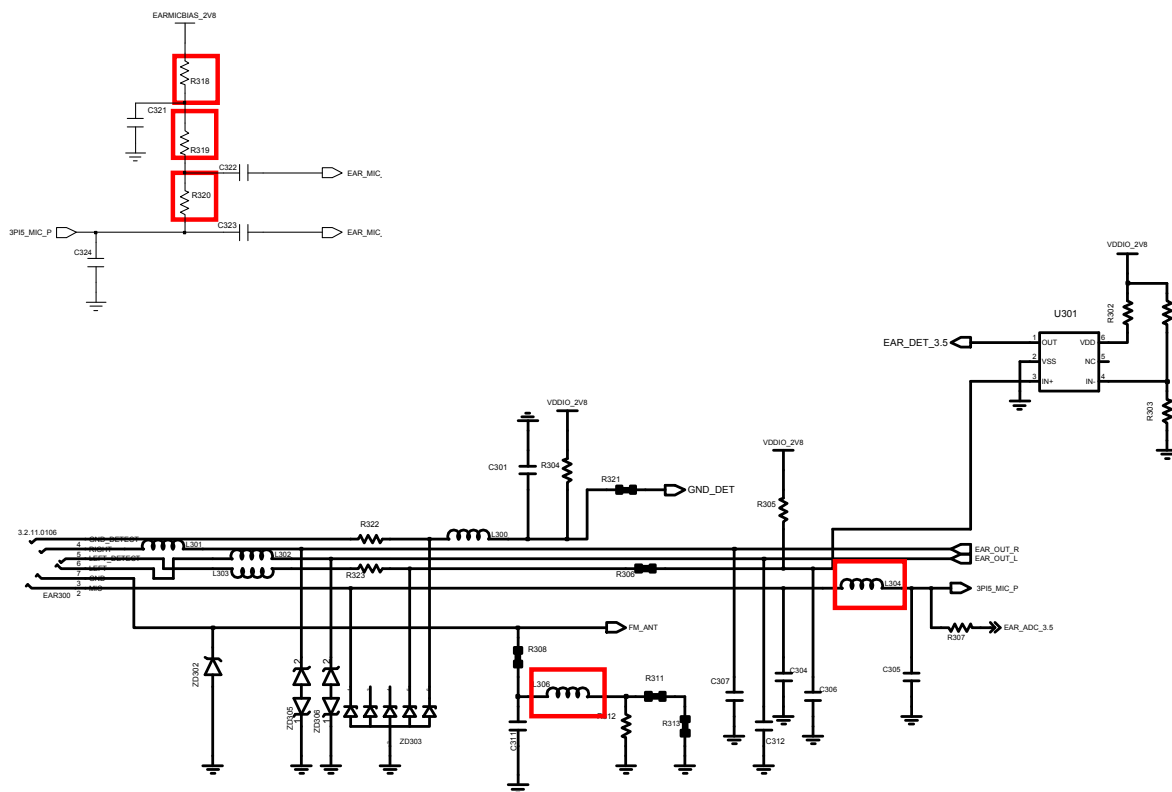


SPEAKER

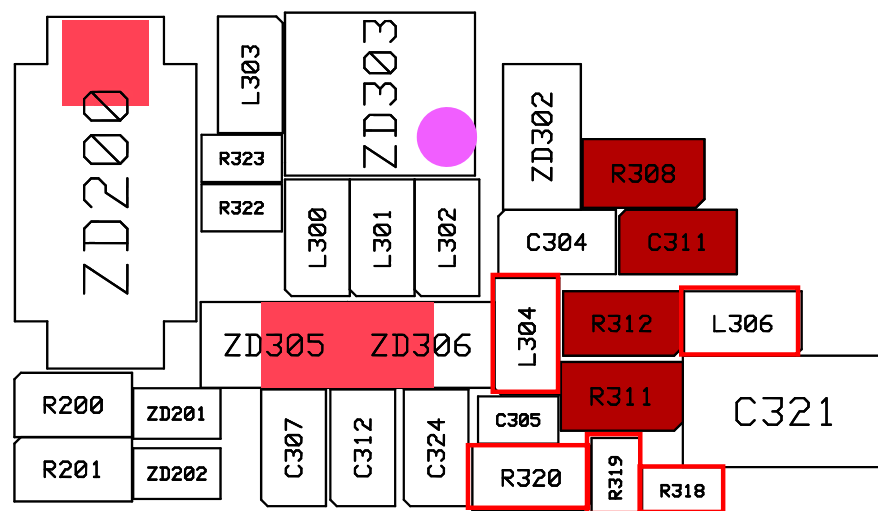


8-3-7. Ear Microphone Part

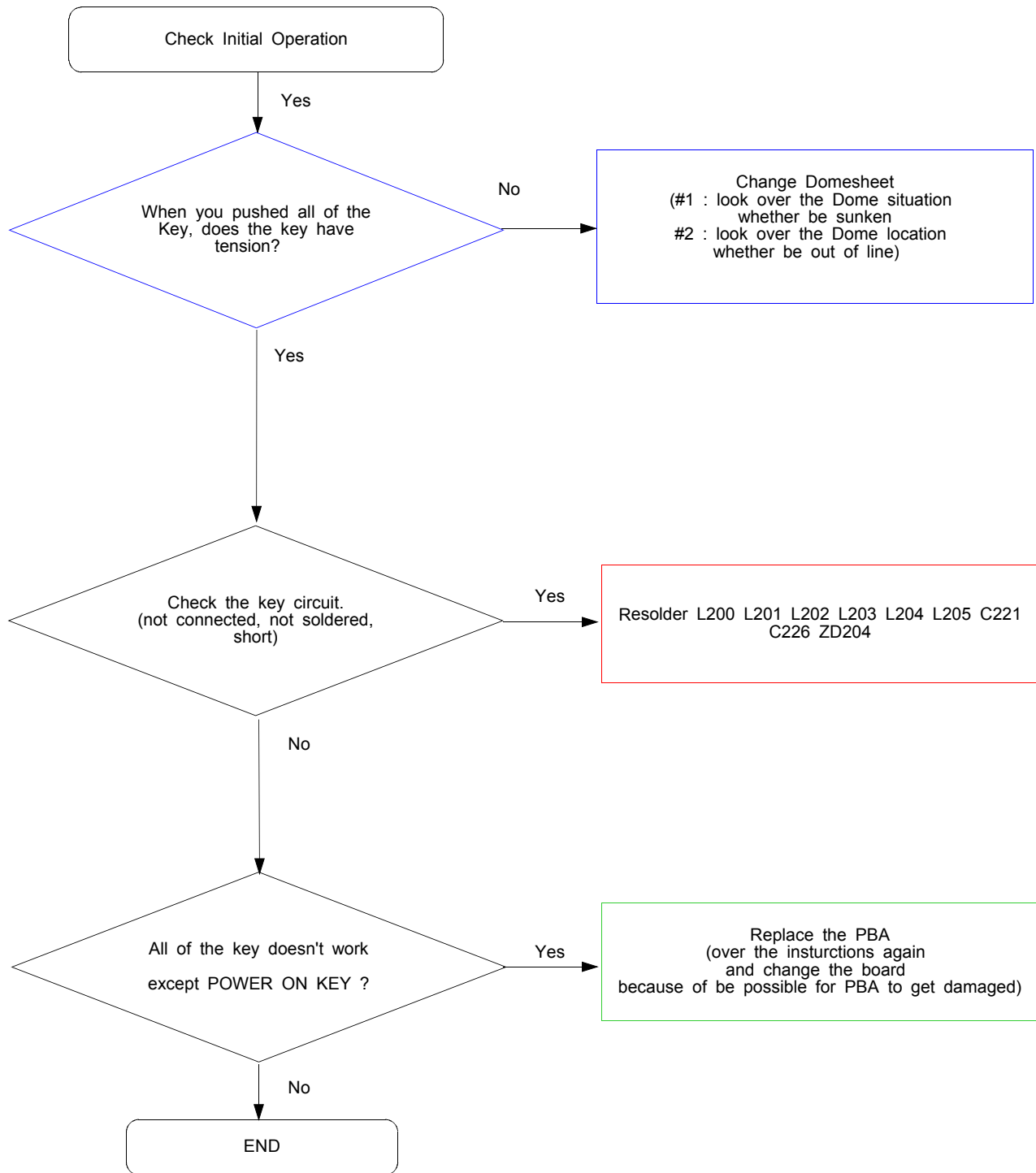


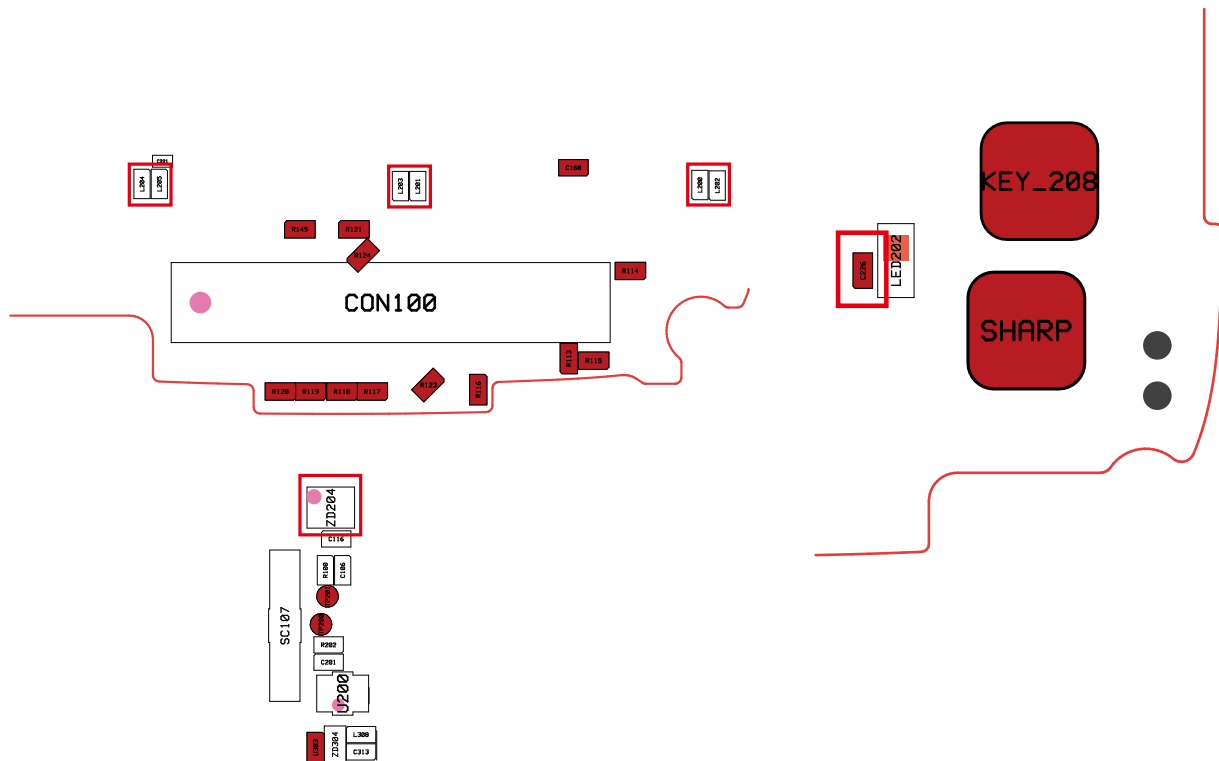
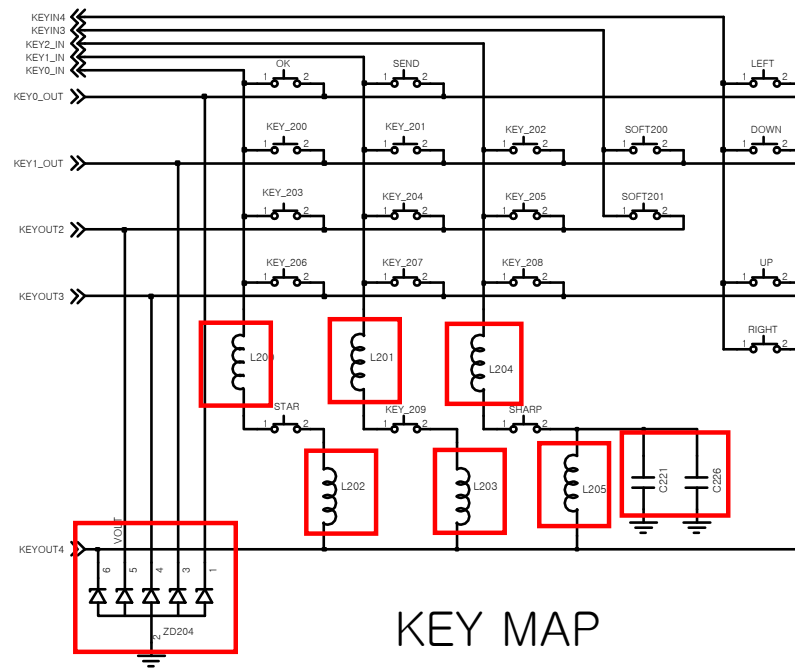


3.5 pi EAR JACK

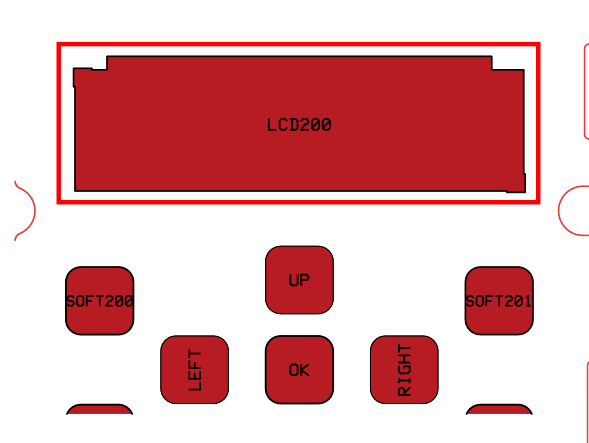
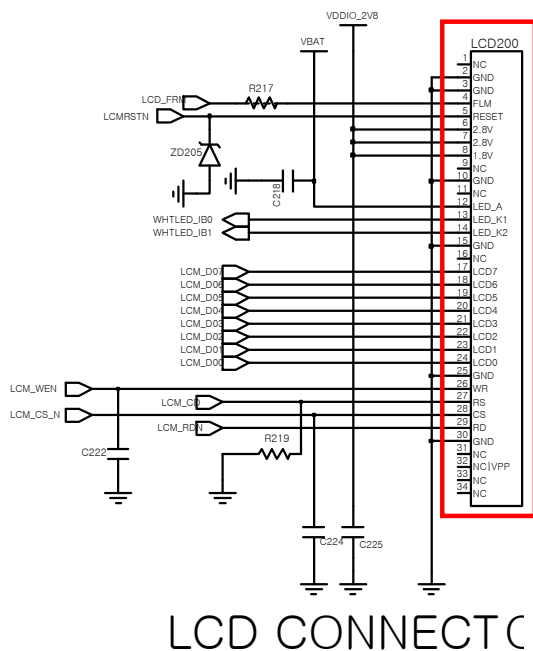
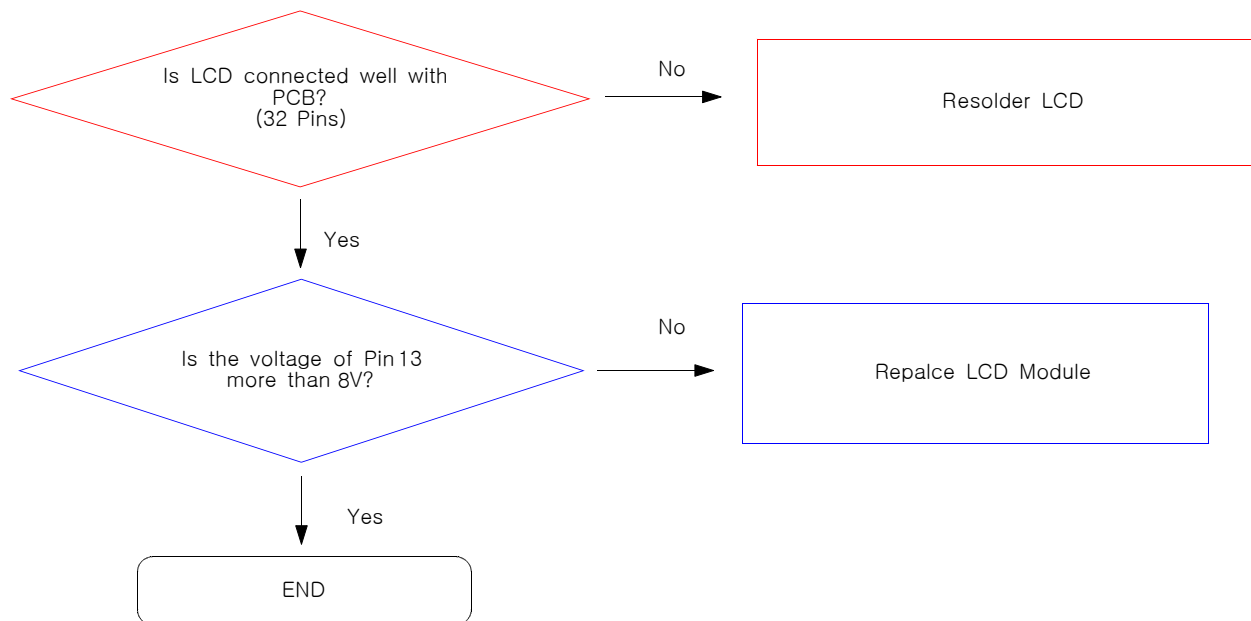


8-3-8. Key Data Input

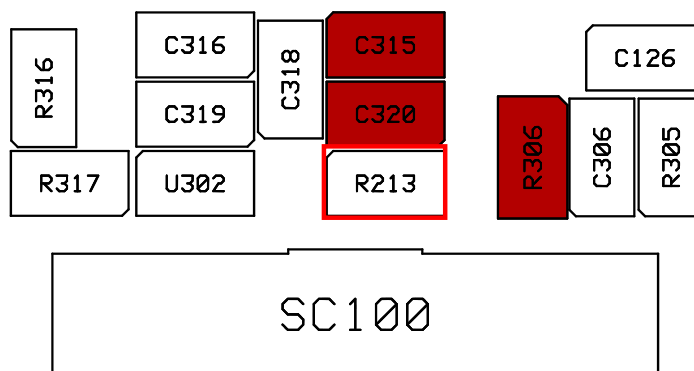
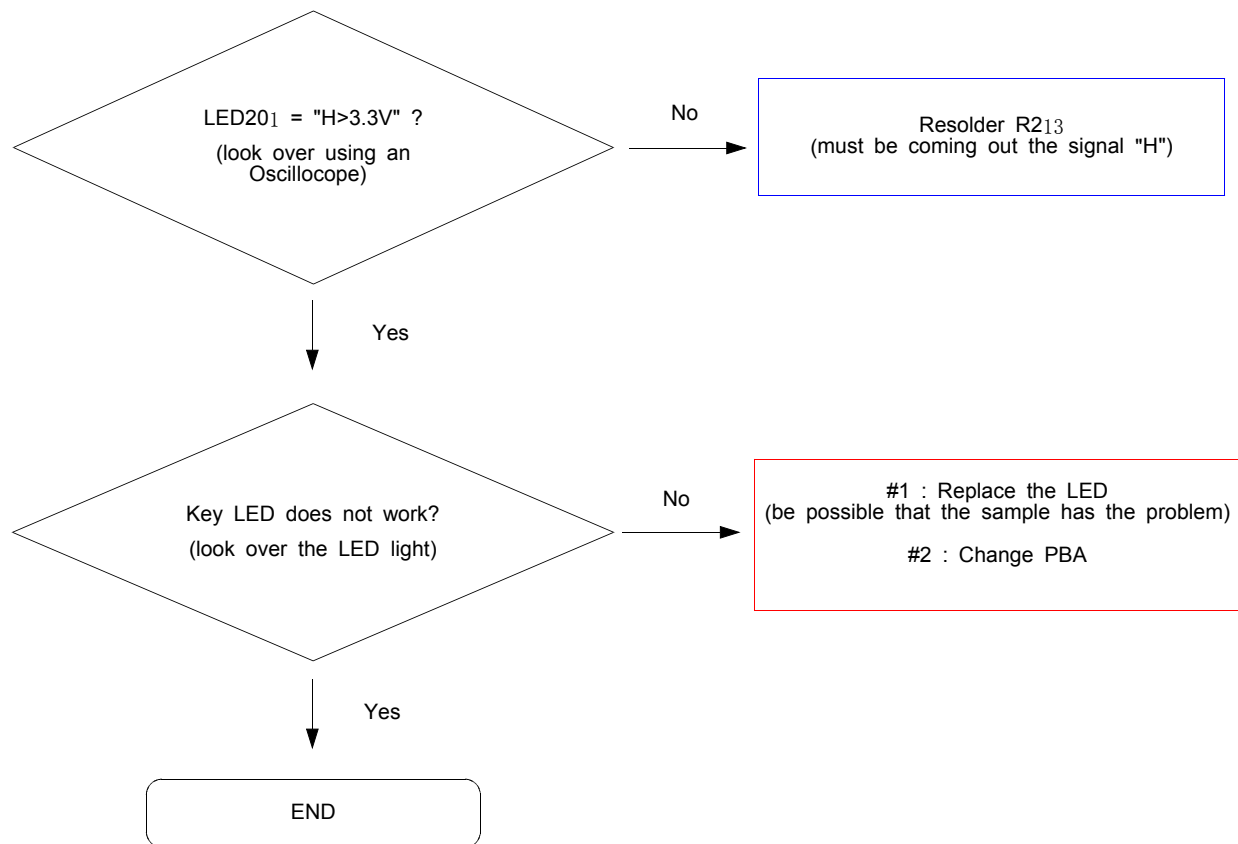


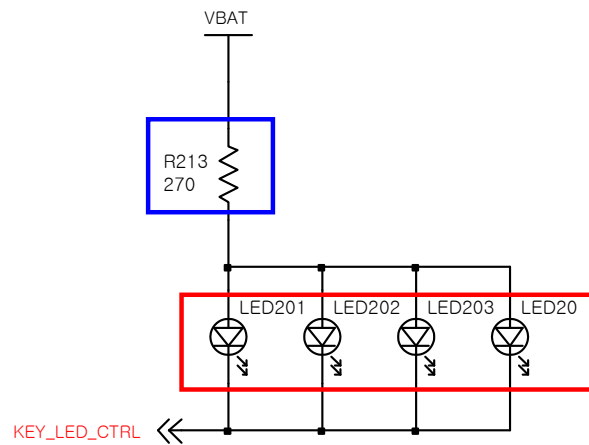


8-39. Back Light (for Color Main LCD)

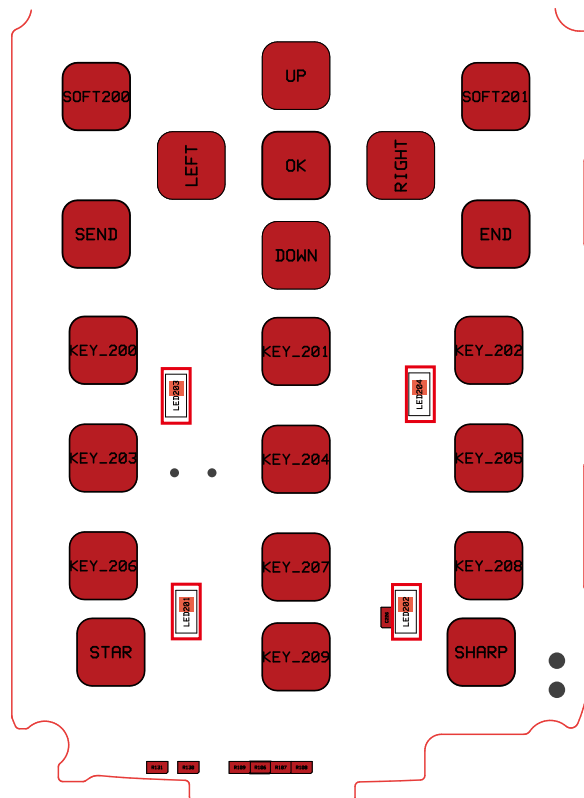


8-3-10. Key Back Light





KEY LED



※ presetting 8960

(Rx setting)

kim.sa@samsung.com

Call Setup Screen			
Control	Call Setup	Call Params	
Operating Mode	OUT Information	BCH Parameters	
Active Cell	IMSI: Called Num:	Multislot Class (GPRS): Multislot Class (EGPRS):	
Connection Type	Traffic Channel Downlink Power	TCH Parameters	
Auto	Burst 1, 2, 3, 4: Unused Bursts:		
Originate Call	Counters	POTCH Parameters	
Paging IMSI	Page: RACH: PACH: Missing Burst: Corrupt Burst: Decode Error:	OUT IP Tx, Packets: Bytes: OUT IP Rx, Packets: Bytes:	
Handover Setup	Error Reports	Receiver Control	
Cell Info	Burst Timing Error: BLER (Block Error Rate): USF BLER:		
1 of 2	Active Cell Idle Inlier (Offset)	Sys Type: GPRS	

< 8960장비 초기화면 >

1. Active Cell
: select GSM or GPRS
2. Connection Type
: select Auto(GSM), BLER(GPRS)
3. BCH Parameter
: select measuring band (DCS or EGSM)
4. Cell power
: -60dBm

(Tx setting)

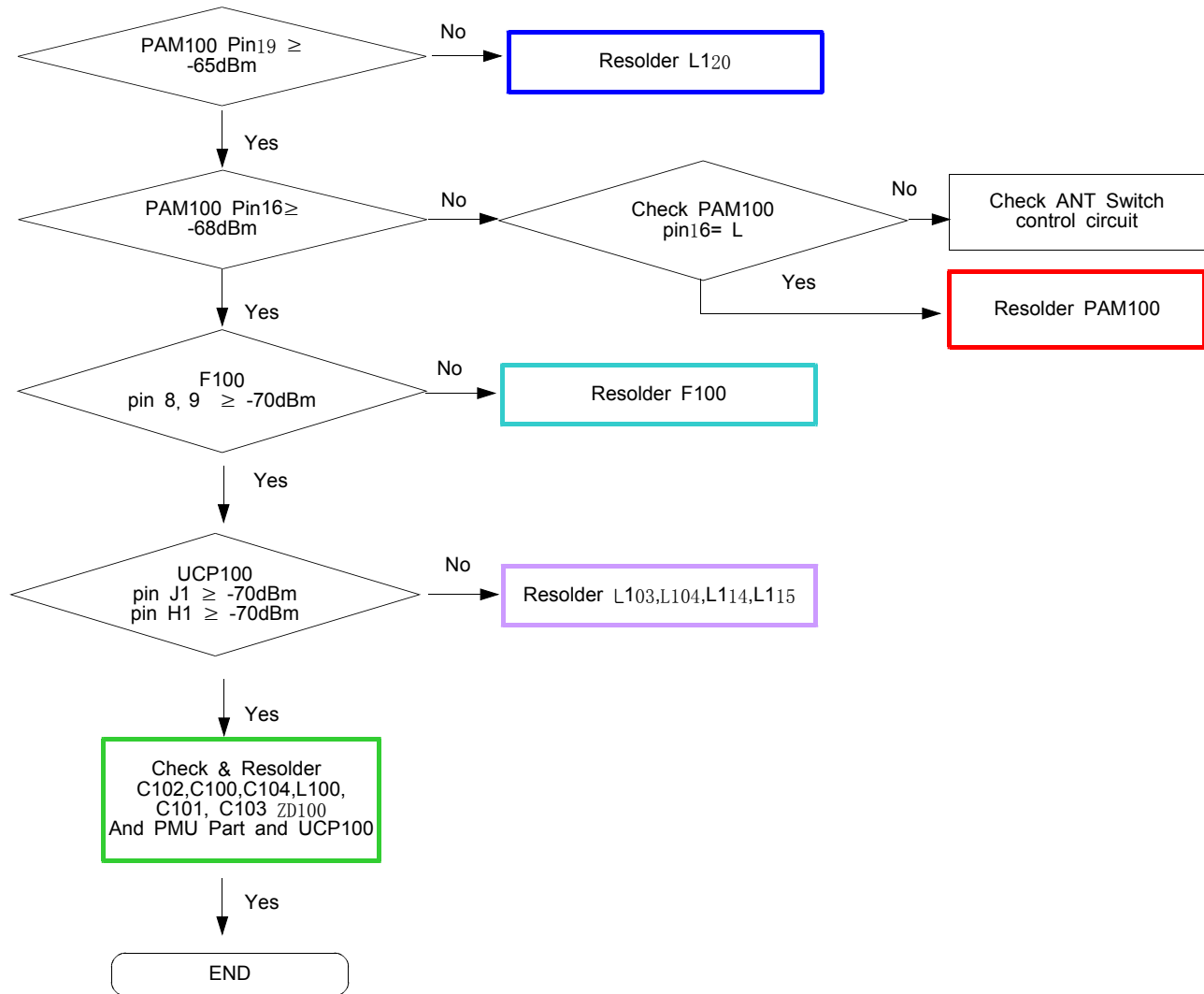
Call Setup Screen			
Control	Call Setup	Call Params	
Operating Mode	OUT Information for IMSI 010405000000640	BCH Parameters	
Active Cell	IMSI: 001010123456789 Called Num: 112	Multislot Class (GPRS): Multislot Class (EGPRS):	
Connection Type	Traffic Channel Downlink Power	TCH Parameters	
Auto	Burst 1, 2, 3, 4: -60.00, Unused Bursts:		
End Call	Counters	POTCH Parameters	
Paging IMSI	Page: 2 RACH: 1 PACH: 0 Missing Burst: 0 Corrupt Burst: 0 Decode Error: 0	OUT IP Tx, Packets: Bytes: OUT IP Rx, Packets: Bytes:	
Handover Setup	Error Reports	Receiver Control	
Cell Info	Burst Timing Error: 0.50 T BLER (Block Error Rate): USF BLER:		
1 of 2	Active Cell Connected Inlier (Offset)	Sys Type: GSM	

< Call이 연결된 화면 >

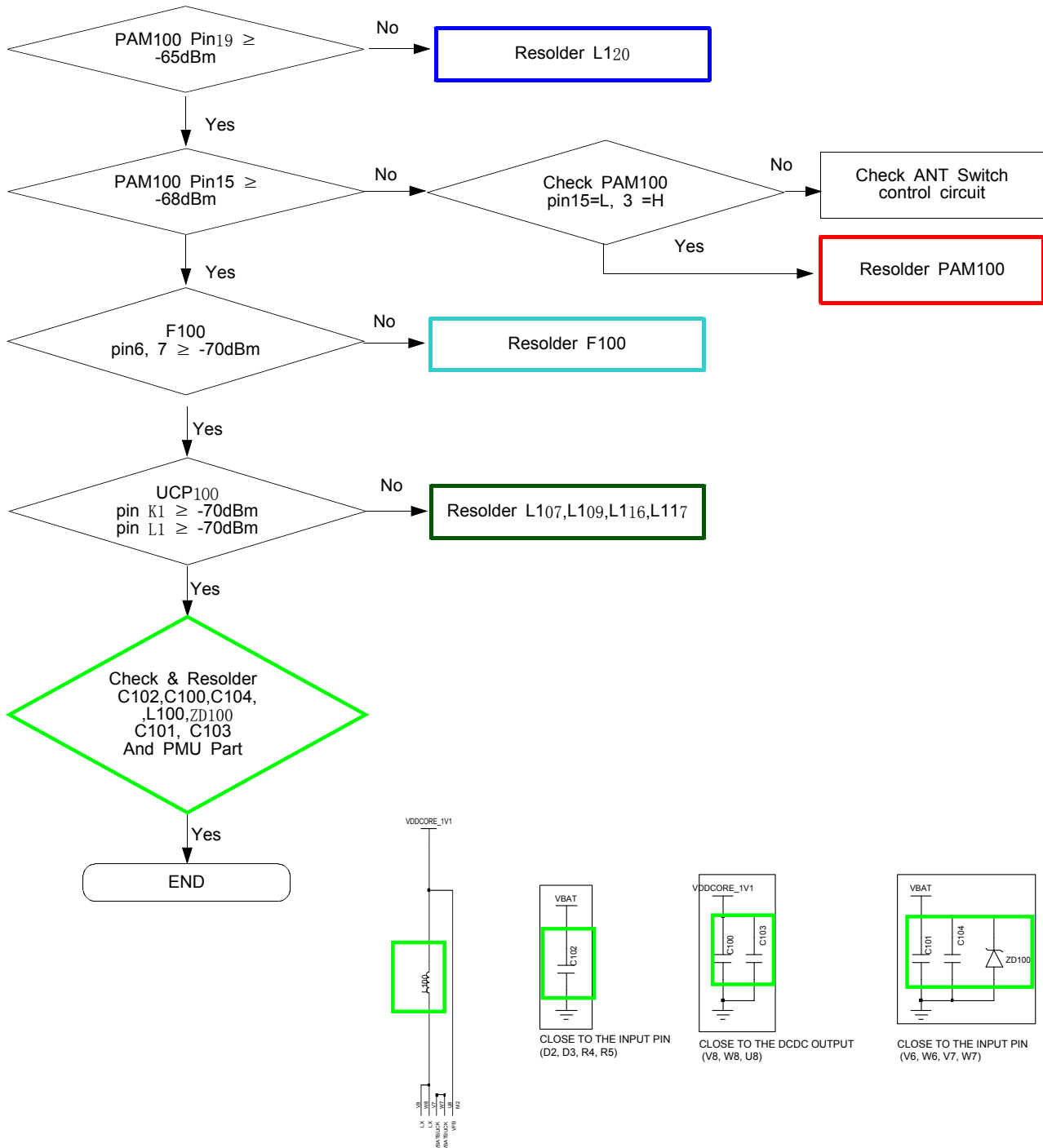
※ After setting 8960 (EGSM / DCS)

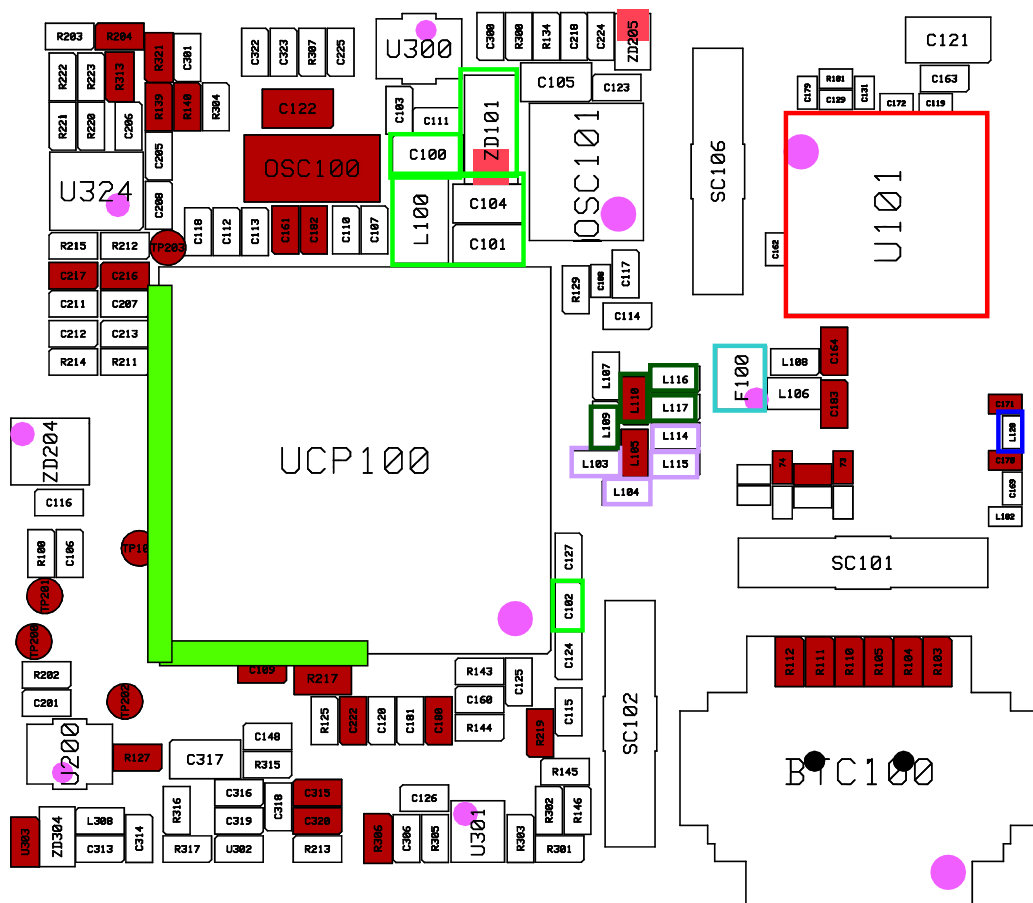
1. After setting, prepare the call setup Display
2. Using an Originate Call, make a call.
3. Confirm the display "connected"
4. start the measuring

8-3-11. GSM900 Receiver

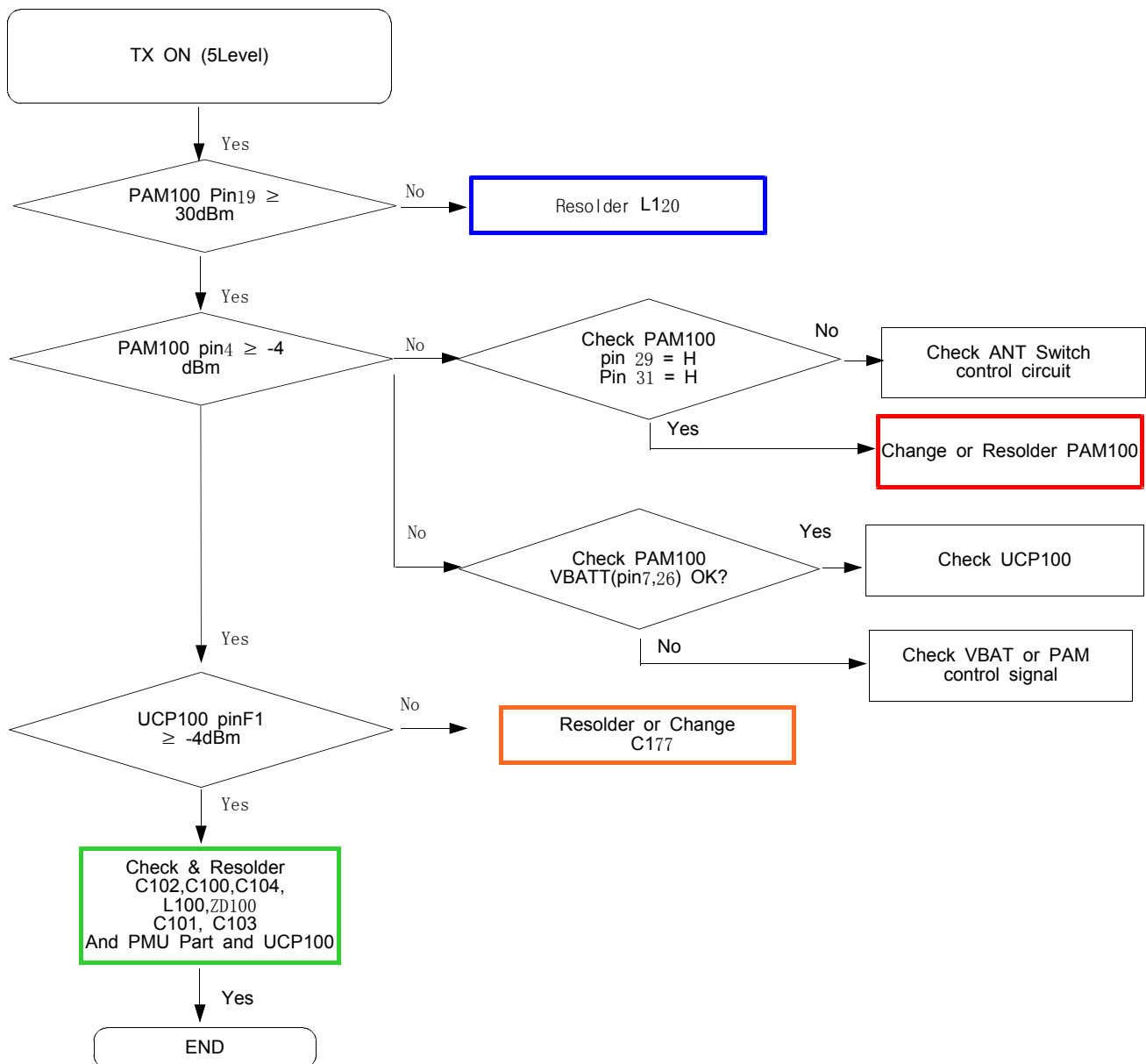


8-3-12. DCS Receiver

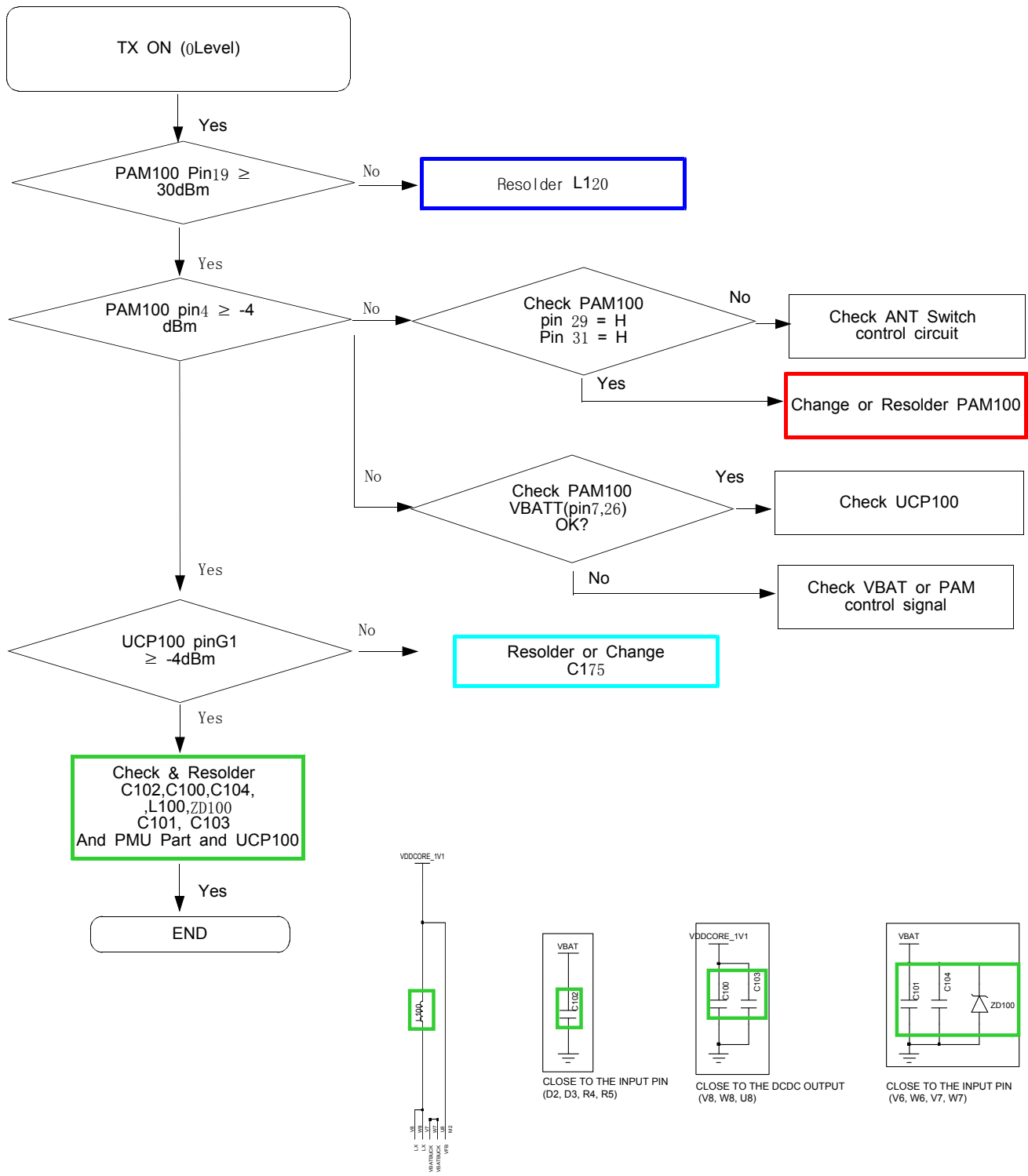


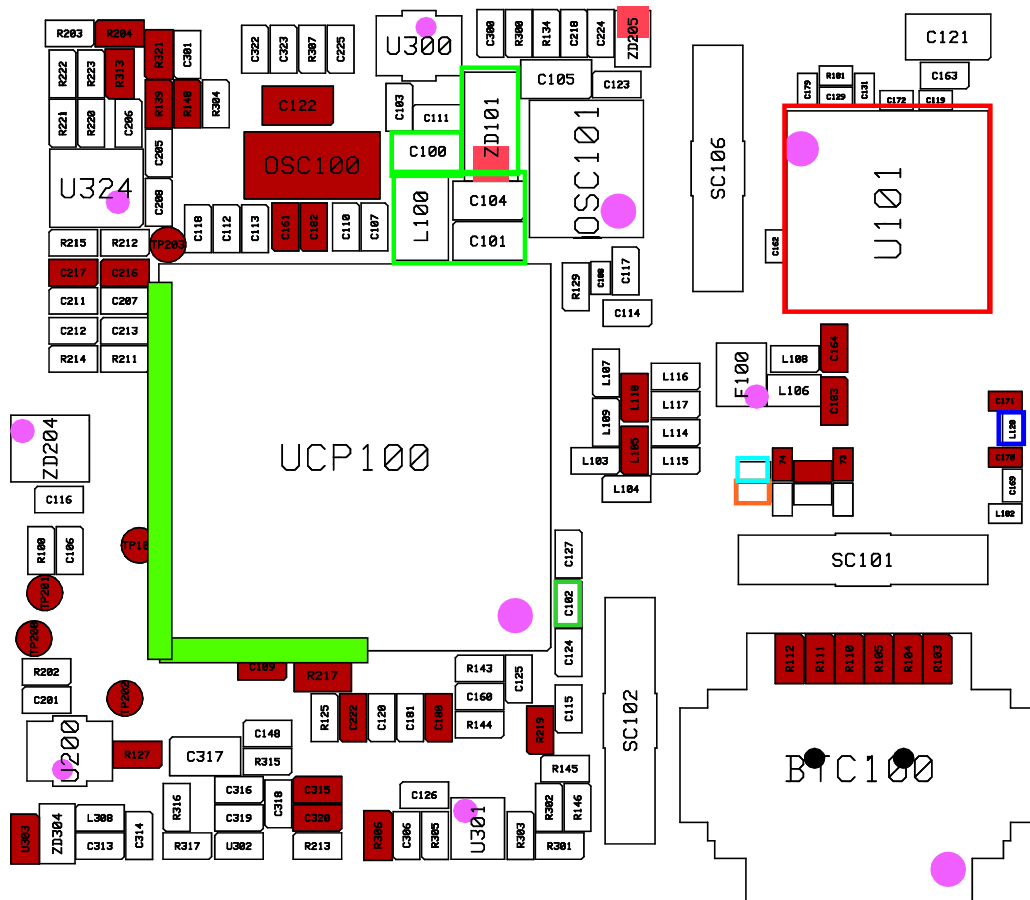
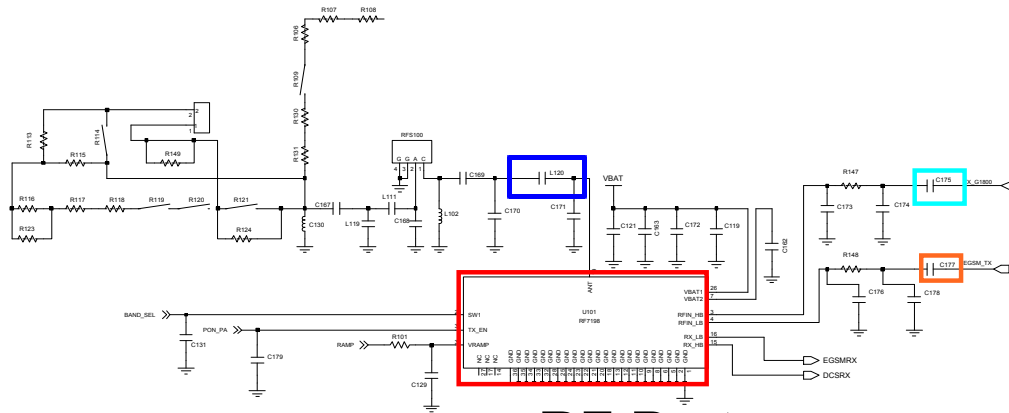


8-3-13. GSM900 Transmitter



8-3-14. DCS Transmitter





8-4. Service Schematics

- Ball LFBGA Pin out(Top View)

UCP100

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
A	VSS RF	VSS RF	VDD 18		VDD SIM0		RF_P AMO DE	RF_P ABA ND		LCM WR		LCM D[6]		LCM D[1]	SIMR ST1		SIMR ST0	VSS	VSS	A
B	VSS RF	AVD DBB	VDD 28	VDD SD	VDD SIM1	TEST RST N	TP_Y U	RF_P AEN	LCM CS1	LCM CD	LCM D[8]	LCM D[4]	LCM D[2]	VLC M	LCM D[0]	SIM DA0	GPIO _63	GPIO _0	VSS	B
C		VME MS	RST N		VSS	PAG SM	TP_Y D	PRO DT	RF_V BS3	LCM CS0	LCM D[7]	LCM D[5]	LCM D[3]	IISC LK	SIMC LK1	SIM DA1	SIMC LK0	IISM CK	IISL RCK	C
D		VBA T	VBA T	VSS	ADCI 0		TP_X L	PBIN T	LDO MEM SR	RF_V BS1	LCM RST N	LCM RD		PWM		EXTI NT0	CLK _AU X	IISD O		D
E		VDDR F			ADCI 2	ADCI 3	TP_X R			RF_V BS2			VSS				IISD1	UOC TS	UOR XD	E
F	US/E GSM TX	VSS RF												VSS		SD_C LK1	PAD P_US R	PAD M_US SR	UOT XD	F
G	DCS/ PCS_ TX	VSS RF				ADCI 1						VDD IO	VDD _D			SD_ D3	UOR TS	VDD USB		G
H	US/E GSM RX		VSS RF			VSS				VSS							SD_C LK	SD_ D2	SD_C MD	H
J	US/E GSM RX	VSS RF			VSS		MIC N	MICP	VSS		VSS						SD_ D0	FSM CSN1	SD_ D1	J
K	DCS/ PCS_ RX					VSS				VSS		VSS			VDD 18_B YP	KEY1 N[6]				K
L	DCS/ PCS_ RX	VSS RF	VDD RF_B VP						HEA D_P_ 1					VSS		KEY OUT[4]	KEY1 N[2]	KEY1 N[5]		L
M		VDD COR E		KP_ OUT	WHT LED_ IR3				HEA D_P_ R		VSS					KEY1 N[1]	KEY1 N[3]		KEY1 N[0]	M
N	VIBR _OU T	VSS RF			WHTL ED_ IB0					VSS				VSS		KEY OUT[6]	KEY1 N[4]		KEY1 N[7]	N
P		DVDD 18	WHT LED_ IR2	WHT LED_ IR1													KEY OUT[2]		KEY OUT[5]	P
R	WHT LED_ RSET	AVD DBB		VBA T	VBA T				EAR N						VSS		KEY OUT[0]	KEY OUT[3]		R
T	REF_ IN-		VDR V	VSSP A		VSS DCD C			EAR P	RTC3 2KO	MIC BIAS						KEY OUT[1]	FSM D3		T
U	REF_ IN+	REF1 _OU T	VCH G	VSSP A		VSS DCD C		VFB		RTC3 2KI	AIR1	AIL1	HEA DMI CP		MTC K		FSM CLK	FSM D0		U
V	VSS RF	ISEN SE	VBA TSEN	OUT PPA	OUT NPA	VBA TPA	VBA TBU CK	LX	VDD RTC		VBA TBK	HEA DMI C_IN	HEA DMI CN	MTD I	MTR STN	FSM CSN0	FSM D2	FSM D1	VSS	V
W	VSS RF	VSS RF	VDD PA	OUT PPA	OUT NPA	VBA TPA	VBA TBU CK	LX	VDD AO		VCO M	AVD DVB	VDD 25	MTM S	MTD O			VSS	VSS	W
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	